

Lesson 10

Chapter 3 (continued): welfare evaluation — equivalent variation, compensating variation and consumer surplus

1 The money-metric welfare change

Revisiting the money-metric indirect utility function to derive the welfare impact of a price change from p^0 to p' , given the benchmark price vector $\bar{p}$:

$$v_m(p', w | \bar{p}) - v_m(p^0, w | \bar{p}) = e(\bar{p}, v(p', w)) - e(\bar{p}, v(p^0, w)).$$

If positive, then the consumer is better off under p' than under p^0 .

The *sign* is the same regardless of $\bar{p}$, but the *size* does depend on $\bar{p}$, simply because we are using different prices to evaluate the welfare impact.

Two benchmarks are particularly natural: $\bar{p} = p^0$ and $\bar{p} = p'$, i.e. evaluating the change at the old prices or at the new prices.

Throughout, write $u^0 = v(p^0, w)$ and $u' = v(p', w)$ for the utility levels before and after the price change. By duality, $e(p^0, u^0) = w = e(p', u')$.

2 Equivalent and compensating variation

Equivalent variation ($\bar{p} = p^0$)

If $\bar{p} = p^0$, the equivalent variation is

$$\begin{aligned} EV(p^0, p', w) &= e(p^0, v(p', w)) - e(p^0, v(p^0, w)) \\ &= e(p^0, u') - e(p^0, u^0) \\ &= e(p^0, u') - w. \end{aligned}$$

If, under p^0 , the consumer receives $e(p^0, u') - w$, then he can afford u' , because his total wealth is $e(p^0, u')$. So this is the amount the consumer would have to be paid at the old prices p^0 to be willing to forgo the price change. Hence it is the amount that is *equivalent* to the price change.

Compensating variation ($\bar{p} = p'$)

If $\bar{p} = p'$, the compensating variation is

$$\begin{aligned} CV(p^0, p', w) &= e(p', v(p', w)) - e(p', v(p^0, w)) \\ &= e(p', u') - e(p', u^0) \\ &= w - e(p', u^0). \end{aligned}$$

If, at the new prices p' , you give up $w - e(p', u^0)$, you end up with wealth $e(p', u^0)$, which is just enough to get you back to the old level of utility u^0 . It is the amount needed to *compensate* for the change in prices.

Example. $p'_i < p_i^0$, all else equal. The agent is better off: $u' > u^0$. Both EV and CV have to be positive, but they are likely to be different.

3 Comparing EV and CV

Let's try to compare the size of EV and CV . From Shephard's lemma, there is a link between the expenditure function and Hicksian demand. Let us use that link, together with the link between Hicksian and Marshallian demand in the Slutsky equation.

Recall

$$\begin{aligned} EV(p^0, p', w) &= e(p^0, u') - w \\ &= e(p^0, u') - e(p', u'). \end{aligned}$$

If only p_i changes, then using the fundamental theorem of calculus,

$$EV(p^0, p', w) = \int_{p'_i}^{p_i^0} \frac{\partial e(p, u')}{\partial p_i} dp_i = \int_{p'_i}^{p_i^0} h_i(p_1^0, p_2^0, \dots, p_i, \dots, p_L^0, u') dp_i.$$

$\Rightarrow EV$ = area under the Hicksian demand curve evaluated at u' ;

CV = area under the Hicksian demand curve evaluated at u^0 .

We need to compare Hicksian demand at different utility levels. We take a few steps.

Step 1. Hicksian vs. Marshallian demand for fixed u

The Slutsky equation, evaluated at $u = v(p, w)$ (equivalently $w = e(p, u)$):

$$\frac{\partial h_i(p, u)}{\partial p_i} = \frac{\partial x_i(p, w)}{\partial p_i} + \frac{\partial x_i(p, w)}{\partial w} x_i(p, w) \leq 0,$$

because of concavity of the expenditure function: the negative own-price substitution effect — Hicksian demand slopes down.

At this point, add the assumption that good i is a *normal* good:

$$\frac{\partial x_i(p, w)}{\partial w} > 0 \quad \Rightarrow \quad \frac{\partial x_i(p, w)}{\partial p_i} < 0,$$

so Marshallian demand slopes down: it is an *ordinary* good. Moreover, at the same point ($u = v(p, w)$) and whenever $x_i(p, w) > 0$,

$$\frac{\partial h_i(p, u)}{\partial p_i} > \frac{\partial x_i(p, w)}{\partial p_i}.$$

The left-hand side is less negative than the right-hand side.

$\Rightarrow$ Hicksian demand is less responsive to its own price than Marshallian demand;

$\Rightarrow$ in the usual graph, with price on the vertical axis (the inverse demand curve), the Hicksian curve is steeper.

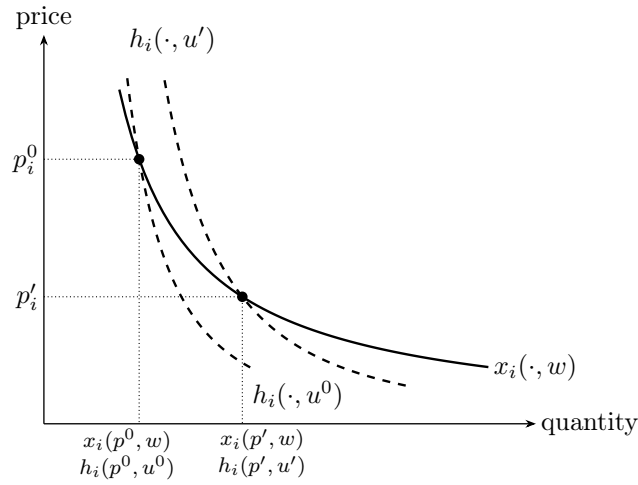
Step 2. Where the curves cross

An implication is that Hicksian and Marshallian demand can cross only once (with the other prices fixed at p_{-i}^0 and w fixed). What can we say about the price at which the crossing occurs, and how does it depend on utility?

For concreteness, assume $p'_i < p_i^0$, and thus $u' > u^0$. From duality,

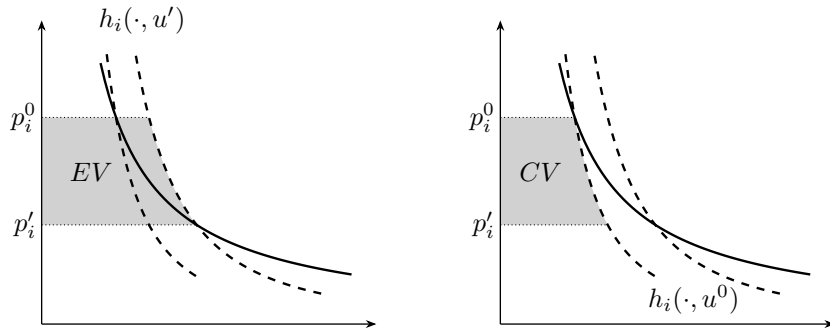
$$\begin{aligned} h_i(p^0, u^0) &= x_i(p^0, e(p^0, u^0)) = x_i(p^0, w), \\ h_i(p', u') &= x_i(p', e(p', u')) = x_i(p', w) > x_i(p^0, w), \end{aligned}$$

since the good is an ordinary good. So $h_i(\cdot, u^0)$ crosses Marshallian demand at p_i^0 , and $h_i(\cdot, u')$ crosses it at p'_i .



Step 3. Put everything together: compare EV and CV

Both are areas under Hicksian demand, between p'_i and p_i^0 : EV under $h_i(\cdot, u')$, CV under $h_i(\cdot, u^0)$.



Since $h_i(\cdot, u')$ lies to the right of $h_i(\cdot, u^0)$,

$$EV > CV.$$

EV and CV both measure the change in indirect utility, but in different ways; both are sound concepts.

4 Consumer surplus

If you used the area under Marshallian demand — consumer surplus — you get neither EV nor CV . Conceptually: what is its meaning?

CS is on shaky ground. The reason is that wealth effects imply that Marshallian demand $\neq$ Hicksian demand.

The problem disappears if there are no wealth effects. Quasilinear preferences have this property — no wealth effect:

$$u(x_i, x_j) = x_j + \varphi(x_i).$$

Indifference curves move in a parallel way, and a change in p_i has no wealth effect on the consumption of x_i (at an interior solution).

In this case $EV = CV = CS$, because the three curves coincide.

Transcription notes

- Notation follows Lessons 7 and 9: the original's capitals $P, E, H, X, V(P, w)$ are typeset as $p, e, h, x, v(p, w)$. The original's utility levels V^0, V' are typeset u^0, u' , so that v stays the indirect utility *function* (as $\bar{u}$ in Lesson 9). EV, CV and CS keep their capitals. **Added:** the line defining u^0, u' and the duality identities $e(p^0, u^0) = w = e(p', u')$ that the derivations use.
- **Clarified:** the original says “today” both for the old prices (“paid today to forgo the price change”) and for the new ones (“under p' today”), and calls both benchmarks “today’s price”; each is replaced by *at the old prices p^0 or at the new prices p'* . **Added:** the Slutsky equation and the slope comparison hold at $u = v(p, w)$, the comparison strictly only when $x_i > 0$. “Hicksian demand is flatter $\Rightarrow$ the inverse hicksian demand curve is steeper” is typeset as *less responsive to its own price / steeper with price on the vertical axis*. Curves in the figures, written $H_i(p, V')$, $x_i(p, w)$, are labelled $h_i(\cdot, u')$, $x_i(\cdot, w)$: they are drawn against p_i with the other prices held at p_{-i}^0 .
- The title page reads “Class 10” only; the subtitle is written to match Lesson 9. No date is given.
- Page 2: “Visiting the money metric indirect utility function to derive of the welfare impact” $\rightarrow$ *Revisiting ... to derive the welfare impact*; “price Δ ” $\rightarrow$ *price change*. “the size does depend on $\bar{p}$ ” — the prime is read as a stray mark ($\bar{p}$). “weafare” $\rightarrow$ *welfare*; “equolent variation” $\rightarrow$ *equivalent variation*.
- Page 3: “the consume receives ... Then he can afford. until v' ” $\rightarrow$ *then he can afford u'* ; “amout” $\rightarrow$ *amount*. In the CV text, “If under p' today $w - E(p', v^0)$ ” has no verb; read as *you give up $w - E(p', v^0)$* (the amount is positive when $u' > u^0$). “tou” $\rightarrow$ *you*. The CV definition is written with $\bar{p} - p'$ for $\bar{p} = p'$, and its second line ends $E(p', v^0)$, typeset as written.
- Page 4: “EV and CV ... But likely to be diff.” $\rightarrow$ *different*. “fundemental of calculee” $\rightarrow$ *fundamental theorem of calculus*. The integrand is written $H_i(P_1^0, P_2^0, \dots, P_i, \dots, P_i^0, V')$; the last price is read as p_L^0 (only p_i varies).
- Page 5: “EV = are under hicksian demand curve evaluation” $\rightarrow$ *area ... evaluated at u'* ; “CV = ... v^0 ” is a ditto line. The Slutsky equation’s left side is written $\partial H_i(P, u)$; typeset $h_i(p, u)$. “because concavity” is completed as *concavity of the expenditure function* (Lesson 9).
- **Corrected:** “add assumption that good i is a common good $\partial X_i(P, U)/\partial U > 0$ ”. The property used in the Slutsky equation is $\partial x_i(p, w)/\partial w > 0$, i.e. a *normal* good; typeset that way. **Added:** the inequality $\partial h_i/\partial p_i > \partial x_i/\partial p_i$ is strict only when $x_i > 0$ (it follows by subtracting the positive wealth term), and the parenthetical “price on the vertical axis” explaining why “flatter” and “steeper” are both right.
- Page 7: the second duality line is written $x_i(p', E(p', u'))$; read as $e(p', u')$. “Since good is an ordinary good” $\rightarrow$ *the good*. **Added:** the sentence stating where each Hicksian curve crosses Marshallian demand.

- Figures: the original is hand-drawn in red. Page 7 shows Marshallian demand $x_i(p, w)$ and two steeper curves, the right one labelled $H_i(p, v')$ and the left one unlabelled; the left one is labelled $h_i(\cdot, u^0)$ here, as the axis labels $H_i(p^0, v^0)$ require. Page 8 shows two panels with the area left of the Hicksian curve between the two price lines hatched in blue, labelled EV (left) and CV (right). They are redrawn with illustrative curves ($p = 4/x$, $p = 4.8/x^2$, $p = 10/x^2$), grey fill for the hatching and dashed lines for Hicksian demand.
- Page 8: “Both = area under hicksian demand” kept; “both measures change in indirect utility but in different ways sound concepts” $\rightarrow \dots$; *both are sound concepts*.
- Page 9: “you get either EV nor CV ” $\rightarrow$ *neither*; “conceptualization: what’s the meanuy?” $\rightarrow$ *Conceptually: what is its meaning?*; “disapears” $\rightarrow$ *disappears*; “this properties” $\rightarrow$ *this property*.
- Page 10: the quasilinear utility is written $V(x_i, x_j) = x_j + \hat{V}(x_i)$; typeset $u(x_i, x_j) = x_j + \varphi(x_i)$, renamed so that it does not clash with the indirect utility function v . **Corrected:** “a Δ in p_i does not impact consumption of x_i ” — with quasilinear utility a change in p_i *does* change x_i (through the substitution effect); what vanishes is the wealth effect, so the line is typeset as “has no wealth effect on the consumption of x_i ”. The closing “(WHEN intension)” is not legible as written; it is read as *at an interior solution [unclear: possible reading]*.
- Pages of the original, for tracing back: page 1 title; page 2 the money-metric welfare change, the two natural benchmarks and the start of EV ; page 3 EV and its interpretation, CV ; page 4 the example, the Shephard’s-lemma route and the EV integral; page 5 EV/CV as areas, step 1 (Slutsky equation, normal good); page 6 Hicksian flatter, step 2; page 7 duality crossings and the demand-curve graph; page 8 the EV/CV panels and $EV > CV$; page 9 consumer surplus and quasilinearity; page 10 the quasilinear utility function.